CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 3: Project Design Phase

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 3: Project Design Phase**
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Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. System Architecture: Snowflake Medallion Pattern

OrderPulse adopts the industry-standard **Medallion Data Architecture** inside Snowflake, establishing clear physical separation of data stages from landing to analytical consumption:

Layer	Schema Name	Key Responsibilities & Design Patterns
RAW (Bronze)	**ORDERPULSE_DB.RAW**	Landing tables with permissive VARCHAR columns. Direct ingestion from @ORDERPULSE_STAGE via COPY INTO. Preserves immutable source payload.
STAGING (Silver)	**ORDERPULSE_DB.STAGING**	Strict data typing (DATE, TIMESTAMP_NTZ, NUMBER), uppercase/casing hygiene, null imputation, deduplication, and operational flag generation (is_realized, is_delivered_late).
CURATED (Gold)	**ORDERPULSE_DB.CURATED**	Star Schema dimensional tables and fact tables linked via MD5 surrogate keys. Hosts pre-aggregated analytical views (V_PRIYA, V_DANIEL, V_GRACE) for reporting.
CONSUMPTION	**Streamlit Web Application**	Interactive 5-tab dashboard fetching curated metrics via Python Snowflake Connector or SQLite local fallback engine.

2. Dimensional Data Model: Star Schema Design

The curated layer models the business using an optimized **Star Schema** architecture designed for fast aggregation and drill-downs:

Table Name	Type & Grain	Keys (PK / FK)	Engineered Analytical Attributes
FACT_ORDERS	Fact (1 Order)	PK: order_key FK: customer_key, restaurant_key, date_key, time_key	gross_order_value, realized_revenue, discount_amount, delivery_time_mins, basket_item_quantity, is_delivered, is_delivered_late, is_cancelled
FACT_ORDER_ITEMS	Fact (1 Line Item)	PK: order_item_key FK: order_key, item_key	quantity, item_unit_price, item_total_price
DIM_CUSTOMERS	Dimension (1 Customer)	PK: customer_key	customer_id, full_name, city, total_orders, lifetime_spend, loyalty_segment (VIP/Frequent/Occasional/Single), is_repeat_customer, customer_aov
DIM_RESTAURANTS	Dimension (1 Restaurant)	PK: restaurant_key	restaurant_id, name, city, cuisine, average_rating, rating_tier (Top/High/Avg/Low), cost_tier, is_active
DIM_MENU_ITEMS	Dimension (1 Dish)	PK: item_key	item_id, item_name, category, price, is_veg, price_tier (Budget/Mid/Premium)

DIM_DATE	Dimension (1 Day)	PK: date_key	calendar_date, year, quarter, month, month_name, day_of_month, day_of_week, is_weekend
DIM_TIME	Dimension (1 Minute)	PK: time_key	time_of_day, hour, minute, meal_period (Breakfast, Lunch Peak, Snacks, Dinner Peak, Late Night)

3. Persona Data Mart Views

- **V_PRIYA_CUSTOMER_ENGAGEMENT:** Aggregates loyalty tiers, repeat purchase rates (69.8%), and order frequency across operational cities to drive retention campaigns.
- **V_DANIEL_RESTAURANT_PARTNERSHIPS:** Cross-tabulates restaurant star ratings against order volumes and GMV, providing statistical evidence that 4.5+ star ratings yield +28% order volume.
- **V_GRACE_EXECUTIVE_OVERVIEW:** Tracks monthly GMV vs. realized revenue, platform delivery SLA compliance rates (81.93% on-time), and regional cancellation root causes.